

HUPSEL, Netherlands

WMO: 062830

Lat: 52.0678N

Lon: 6.6567E

Elev: 29

StdP: 100.98

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -7.4	(c) -5.2	(d) -10.6	(e) 1.5	(f) -5.7	(g) -8.2	(h) 1.9	(i) -4.5	(j) 11.8	(k) 8.9	(l) 10.4	(m) 8.2	(n) 2.8	(o) 60	(p) 0.536

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.9	(c) 29.5	(d) 20.0	(e) 27.1	(f) 19.2	(g) 25.1	(h) 18.4	(i) 21.2	(j) 27.2	(k) 20.2	(l) 25.5	(m) 19.2	(n) 23.7	(o) 3.1	(p) 100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.2	14.0	23.5	18.3	13.2	22.3	17.5	12.6	21.3	61.4	27.1	58.2	25.7	54.9	24.0	25.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.2	8.0	7.0	-9.5	33.9	3.5	2.3	-12.0	35.5	-14.0	36.9	-15.9	38.2	-18.5	39.9		
			-9.9	23.6	3.4	1.0	-12.3	24.3	-14.3	24.9	-16.3	25.5	-18.8	26.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.2	2.6	3.5	6.0	9.5	13.3	16.2	18.0	17.4	14.6	10.8	6.7	3.6		
	DBStd	6.50	4.30	3.86	3.58	3.65	3.65	3.16	3.17	2.79	2.91	3.48	3.50	4.32		
	HDD10.0	951	229	183	131	53	10	0	0	0	1	33	108	202		
	HDD18.3	3047	487	416	383	264	161	79	46	52	117	235	349	458		
	CDD10.0	1028	1	1	6	38	112	186	248	229	139	57	9	2		
	CDD18.3	84	0	0	0	0	5	16	35	23	5	1	0	0		
	CDH23.3	871	0	0	0	10	66	174	357	213	47	2	0	0		
	CDH26.7	246	0	0	0	0	12	45	118	63	8	0	0	0		
(14) Wind	WSAvg	3.3	4.1	4.0	3.8	3.3	3.3	3.0	2.8	2.6	2.7	3.2	3.4	4.0		
(15) Precipitation	PrecAvg	843	73	61	56	46	61	73	85	88	73	70	75	80		
	PrecMax	1133	125	123	127	103	139	170	195	246	179	196	133	180		
	PrecMin	651	5	18	12	0	22	11	30	27	8	22	9	27		
	PrecStd	112	34	30	33	22	27	37	39	53	41	33	31	34		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.6	13.9	18.5	25.1	28.2	31.1	33.5	32.1	27.7	22.1	16.4	13.3		
		MCWB	10.9	9.7	12.1	16.0	18.5	20.5	20.9	21.2	19.7	17.3	13.8	11.5		
	2%	DB	11.0	11.6	15.6	21.5	24.9	27.5	29.7	28.2	24.5	19.0	14.4	12.1		
		MCWB	9.7	9.3	10.9	14.1	16.3	19.0	20.1	20.2	18.7	15.8	12.7	10.8		
	5%	DB	9.7	10.4	13.3	18.8	22.9	25.1	27.1	25.5	22.0	17.0	13.0	10.8		
		MCWB	8.6	8.5	9.9	12.9	15.4	18.3	19.4	19.2	17.5	14.7	11.6	9.6		
	10%	DB	8.4	9.0	11.5	16.4	20.7	22.9	24.8	23.3	19.8	15.7	11.6	9.4		
		MCWB	7.3	7.5	8.9	11.9	14.7	17.0	18.5	17.9	16.3	13.8	10.3	8.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.4	11.1	13.1	16.4	19.6	21.9	22.5	22.6	20.7	17.7	14.1	12.1		
		MCDB	12.1	12.4	16.5	23.2	26.0	29.3	29.4	29.5	25.3	21.0	15.4	12.8		
	2%	WB	10.0	9.8	11.6	14.7	17.8	20.3	21.2	21.0	19.3	16.1	12.9	11.0		
		MCDB	10.9	11.1	14.4	20.0	23.0	25.7	27.6	26.2	23.4	18.6	14.1	12.1		
	5%	WB	8.6	8.7	10.4	13.5	16.4	19.0	20.2	19.8	18.0	15.1	11.7	9.7		
		MCDB	9.6	10.1	12.8	17.8	21.1	23.6	25.8	24.6	21.4	16.8	13.0	10.7		
	10%	WB	7.4	7.5	9.2	12.3	15.2	17.8	19.2	18.6	16.9	14.1	10.5	8.4		
		MCDB	8.3	9.0	11.3	15.9	19.6	21.9	23.8	22.3	19.3	15.6	11.5	9.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.6	5.8	7.8	10.2	10.2	10.2	9.9	9.7	9.1	7.1	5.3	4.4		
		MCDBR	4.9	7.4	11.3	14.2	13.9	14.0	14.7	13.8	12.0	8.8	6.0	4.9		
	5% WB	MCWBR	4.4	5.5	7.4	8.1	6.9	7.2	6.5	6.9	6.9	5.9	4.9	4.3		
		MCWBR	5.0	6.4	9.8	12.6	12.1	12.1	12.6	12.3	10.9	8.2	5.7	4.8		
(40) Clear-Sky Solar Irradiance	taub		0.327	0.349	0.378	0.410	0.413	0.412	0.423	0.415	0.390	0.363	0.341	0.322		
		taud	2.359	2.338	2.267	2.216	2.244	2.266	2.242	2.286	2.345	2.402	2.371	2.340		
	Ebn at Noon		667	759	806	823	839	843	825	810	786	733	639	599		
		Edh at Noon	62	83	107	127	129	128	129	117	99	77	61	54		
(44) All-Sky Solar Radiation	RadAvg		0.70	1.37	2.49	4.03	4.90	5.22	5.00	4.21	3.07	1.80	0.83	0.54		
	RadStd		0.09	0.25	0.35	0.44	0.44	0.42	0.63	0.37	0.36	0.17	0.10	0.06		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (80 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page